

Optimization of waveguide fabrication processes in lithium-niobate-on-insulator platform

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ABSTRACT

Lithium niobate (LN) is used in diverse applications, such as spectroscopy, remote sensing, and quantum communications. The emergence of lithium-niobate-on-insulator (LNOI) technology and its commercial accessibility represent significant milestones. This technology aids in harnessing the full potential of LN's properties, such as achieving tight mode confinement and strong overlap with applied electric fields, which has enabled LNOI-based electro-optic modulators to have ultra-broad bandwidths with low-voltage operation and low power consumption. As a consequence, LNOI devices are emerging as competitive contenders in the integrated photonics landscape. However, the nanofabrication, particularly LN etching, presents a notable challenge. LN is hard, dense, and chemically inert. It has anisotropic etch behavior and a propensity to produce material redeposition during the reactive-ion plasma etch process. These factors make fabricating low-loss LNOI waveguides (WGs) challenging. Recognizing the pivotal role of addressing these fabrication challenges for obtaining low-loss WGs, our research focuses on a systematic study of various process steps in fabricating LNOI WGs and other photonic structures. In particular, our study involves (i) careful selection of hard mask materials, (ii) optimization of inductively coupled plasma etch parameters, and finally, (iii) determining the optimal post-etch cleaning approach to remove redeposited material on the sidewalls of the etched photonic structures. Using the recipe established, we realized optical WGs with a total (propagation and coupling) loss value of -10.5 dB, comparable to established values found in the literature. Our findings broaden our understanding of optimizing fabrication processes for low-loss lithium-niobate WGs and can serve as an accessible resource for advancing the LNOI technology.

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I. INTRODUCTION

Lithium niobate (LiNbO_3 , LN) has attracted considerable attention as a versatile and promising material across various applications. Its significance arises from its exceptional properties, encompassing optical non-linearity, electro-optical, and acousto-optical characteristics, along with a wide optical transparency range (from ≈ 0.35 to ≈ 5.5 μm) and a high refractive index.^{1,2} The successful implementation of periodic poling in LN has given rise to quasi-phase-matched crystals, exemplified by periodically poled LN (PPLN).^{3,4} PPLN crystals have found widespread use in devices such as second harmonic generation,^{5,6} optical parametric oscillators,^{7–11}

spontaneous parametric down-converters,¹² and entangled-photon generators,¹³ finding applications in quantum communication¹⁴ and teleportation of quantum states.¹⁵

Despite widespread use in bulk electro-optic and optical applications, LN strongly lags behind other competing and mature integrated photonic platforms (IPPs), such as silicon (Si) and gallium arsenide (GaAs). This can be attributed to difficulties fabricating low-loss LN waveguides (WGs), integration obstacles, and scalability issues associated with wafer-scale processing.¹⁶ Unlike thin-film silicon-on-insulator and GaAs IPPs, traditional LN WG devices were predominantly fabricated using techniques such as titanium (Ti) in-diffusion or proton exchange in bulk LN.^{17–20} These methods

faced limitations such as weaker mode confinement and large bending radii, thus a large device footprint, weak dispersion engineering, and reduced non-linear conversion efficiencies.²¹ Notwithstanding these challenges, the remarkable properties of lithium niobate position the LN platform as an ideal candidate for a new IPP. The LN IPP combines diverse optical functions on a single chip and can significantly advance the field of integrated photonics.

The landscape has undergone a transformative shift owing to the emergence of high-quality, commercially available lithium-niobate-on-insulator (LNOI) wafers.¹⁶ Capitalizing on this development, and in combination with the rapid progression of scalable micro-/nano-fabrication techniques, a large variety of LNOI photonic devices have been demonstrated. These devices include highly efficient low-loss WGs,^{22–24} non-linear wavelength converters,²⁵ broadband frequency comb sources,²⁶ high-performance modulators,²⁷ and integrated cryogenic detection devices.²⁸ Leveraging tighter confinement and substantial index contrast compared to ion-exchanged LN WGs, LNOI WGs enhance the operational efficiency of devices like electro-optic modulators, enabling operation over an ultra-broad bandwidth with reduced power consumption.²⁹ In addition to these advances, LNOI has the potential to actualize multi-functional, integrated-photonic circuits for both classical and quantum applications.³⁰

A critical factor in realizing LNOI devices is developing a reliable etch recipe capable of transferring the desired patterns of photonic structures from the photo- or electron-beam resist to the LN layer. A reliable LN etch that does not introduce substantial optical losses poses a formidable challenge. Due to its hardness and high density,³¹ LN is difficult to etch using conventional techniques. In addition, its chemical inertness renders it less amenable to chemical etching processes, resulting in an anisotropic etch dictated by its crystallographic orientation.^{32,33} Varying etch rates along different crystal planes result in non-uniform etching rates. A sub-optimal etching process induces significant roughness and leads to non-vertical sidewall angles, thereby contributing to elevated propagation losses in the WGs.¹⁶ These challenges underscore the difficulty of achieving low optical losses in WGs and other photonic structures.

Literature indicates diverse methods employed for LN etching, each with distinct advantages and drawbacks. Diamond-blade dicing, for instance, has proven effective in achieving a high aspect ratio and optically low-loss (<1 dB/cm) ridge WGs.³⁴ However, this technique faces limitations in realizing structures such as directional couplers and bi-layer tapers. Chemical-mechanical polishing lithography has produced sub-nanometer surface roughness in WGs with reported optical losses of 0.027 dB/cm.³⁵ Notwithstanding this achievement, the resulting WGs exhibit shallow sidewalls, imposing constraints on the minimum bending radius, which is essential in micro-ring resonator applications. Alternative methods using wet etching,^{36,37} helium ion implantation,³⁸ and proton exchange³⁹ were also explored. Each was found to have limitations; under-etching is a challenge using the wet etching process, and the latter two methods are affected by issues related to low index contrast, and degradation in electro-optic and nonlinear coefficients, respectively.¹⁶

Reactive-ion plasma etching (RIE) of lithium niobate stands out as a viable and robust approach for fabricating LN WGs. The technique offers the ability to achieve anisotropic etching with

precisely controlled etch depths.¹⁶ Despite its advantages, the process is complex. It demands optimization of numerous etch parameters, including substrate heating, mask material durability during etching, and the mitigation of by-product redeposition during the etch process. In contrast to other IPPs that benefit from well-developed RIE recipes, wherein gases are employed to eliminate redeposited material, LN lacks a comparably robust etch recipe that is well-characterized and easily found in the literature. Previous research explored LN etching using fluorine-based plasma but resulted in the undesired redeposition of lithium fluoride (LiF) on WG sidewalls. LiF presence not only diminishes the etch rate but also introduces heightened sidewall roughness and scattering losses.⁴⁰

Recent low-optical-loss LN WGs have been successfully fabricated through physical etching, specifically milling, utilizing argon (Ar⁺) plasma in an inductively coupled plasma (ICP) dry etch process.^{16,22–24,26,28} Devices fabricated by this method have demonstrated exceptional optical loss of 0.013 dB/cm,²⁴ marking the lowest reported value to date, albeit still exceeding the material absorption rate of 0.002 dB/cm.^{41,42} Challenges using this method include low etch-selectivity and material redeposition during the etch. The former necessitates the incorporation of a hard mask to facilitate deeper etch depths, which is needed for tighter optical-mode confinement.¹⁶ The choice of hard mask determines the sidewall roughness, which impacts the optical losses. Therefore, meticulous consideration in selecting the mask material is imperative to minimize propagation losses.⁴³ The other challenge is material redeposition. Sidewall roughness and optical losses are affected by micro-masking effects from the material redeposited on the sidewalls, and hence, it is essential to obtain redeposit-free sidewalls. In Ref. 32, an optimized wet-chemistry process is employed for redeposition removal. A methodology was recently proposed to remove the redeposit by fine-tuning the ICP parameters during the LN etch process.⁴⁴ In Ref. 44, the tool's direct-current (DC) bias was increased to ≈ 1 kV so that the LN etch rates overcome the redeposition rates, resulting in a regime where the sidewalls are redeposit-free. However, this value of DC bias is not a readily available configuration in the most commonly available ICP RIE etchers. Moreover, the authors of Ref. 44 claim that obtaining these results comes at the cost of maintaining a very sensitive system at a high level, free of contaminants from etching other materials. This is extremely hard to achieve in a research environment where tools are generally shared and used for various processes and materials.

Our work describes a methodology to fabricate optimized and reproducible LN WGs characterized by minimized surface roughness to mitigate light propagation losses. Our recipe was developed at the NIST NanoFab—a shared-use nanofabrication facility. Our approach integrates the best practices with an emphasis on providing essential fabrication tips that, to the best of our knowledge, are not found elsewhere. We show the influence of ICP RIE parameters to control the redeposit on the sidewalls. Most importantly, we produced high-quality LN etch results in a typical user-shared cleanroom. Such a nanofabrication facility focuses on research and development rather than production and, thus, lacks a dedicated ICP RIE tool for LN etch. The ICP RIE tool that we used for the lithium-niobate etch is also used for processing other materials including aluminum (Al), chromium (Cr), chrome oxide (Cr₂O₃),

molybdenum (Mo), tantalum (Ta), and titanium (Ti). The etching of these materials and their various etch chemistries may lead to the presence of different etch residues inside the ICP RIE chamber. Such residues could be difficult to remove during the post-etch chamber cleaning process. As a result, the residues negatively impact the reliability of the LN etch process. Our approach encompasses a comprehensive and systematic fabrication procedure, beginning with (i) investigating the optimal plasma etch chemistry to control the morphology of etched sidewalls and mitigate redeposition, (ii) the careful optimization of appropriate mask materials where we compare soft and hard masks, and (iii) finally, determining an optimal redeposition removal protocol through a wet cleaning process. We anticipate that our findings will provide valuable insights to overcome the multifaceted fabrication challenges associated with developing LNOI-based integrated photonic devices, offering benefits to researchers and the broader scientific community.

Our paper is organized as follows. Section II discusses the fabrication steps in producing LN WGs and structures. Section III discusses the results of this work. In particular, Sec. III A discusses the effect of selecting radio-frequency (RF) bias power. Section III B describes optimization of the redeposition cleaning process. Section III C compares the soft and hard etch masks. Section III D analyzes the LN etch profiles obtained using Cr and SiO₂ hard masks. Finally, Sec. III E presents our optical characterization results. We summarize our findings in Sec. IV.

II. FABRICATION

A 4-in. LNOI wafer with a 700 nm x-cut film, 2 μm SiO₂ insulation layer, and 525 μm thick Si substrate from NANOLN is diced into $10 \times 10 \text{ mm}^2$ chips. We clean the chips with piranha solution ($\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2 = 3:1$), followed by the standard RCA⁴⁵ process, and drying with dry N₂ gas before the lithography step. As mentioned earlier, the intrinsic hardness of LN necessitates using an appropriate

mask material to achieve the desired etch depths. Our work explores three types of masks: soft masks using electron-beam (e-beam) resist, metallic Cr mask, and dielectric SiO₂ hard mask. It should be noted that the physical structure of the LN WGs depends on the mask utilized, a topic that will be explained in Sec. III C. A schematic detailing the fabrication procedure is presented in Fig. 1(a). For the soft-mask samples, the LNOI chips are directly spin-coated with the e-beam resist. In the case of hard mask samples, the appropriate mask material of the requisite thickness is initially deposited onto the chips before the resist spin-coating. For the Cr mask samples, a 50 nm thick film of Cr is deposited using electron-beam deposition. For the SiO₂ hard mask samples, 500 nm of SiO₂ is deposited using plasma-enhanced chemical vapor deposition (PECVD), followed by e-beam deposition of 10 nm of Cr. To avoid the charging effect, the Cr layer is necessary to establish a conduction pathway during the subsequent e-beam lithography (EBL) and LN etching steps.

Our etch-test structures consist of waveguides of 800 nm width and 300 μm length. To define lithographic patterns on all three sets of samples precisely, we use the EBL write tool and a positive-tone ZEP520A e-beam resist. Since we are using a positive-tone resist, our test structures are defined with the help of sleeves, as illustrated in Fig. 2(a). To achieve a one-micron-thick resist layer, we dispense and spin-coat undiluted ZEP520A thrice on each chip, then we soft-bake the chips on a hotplate at 180 $^\circ\text{C}$ for 5 min. The exposed patterns are developed in hexyl acetate at -5°C , followed by rinsing in isopropyl alcohol and distilled water. The developed resist patterns exhibit a sidewall angle of $\sim 90^\circ$. These patterns are transferred to the hard mask layers [Cr (50 nm) and Cr (10 nm)/SiO₂ (500 nm)] using ICP RIE etchers (Oxford PlasmaLab 100 and Oxford PlasmaPro) using the etch chemistries mentioned in Table I. The etch rates for ZEP were 106, 127, and 42.5 nm/min using the Cr, SiO₂, and LN etch recipes, respectively. All three sets of patterned samples are then etched using the ICP RIE tool.

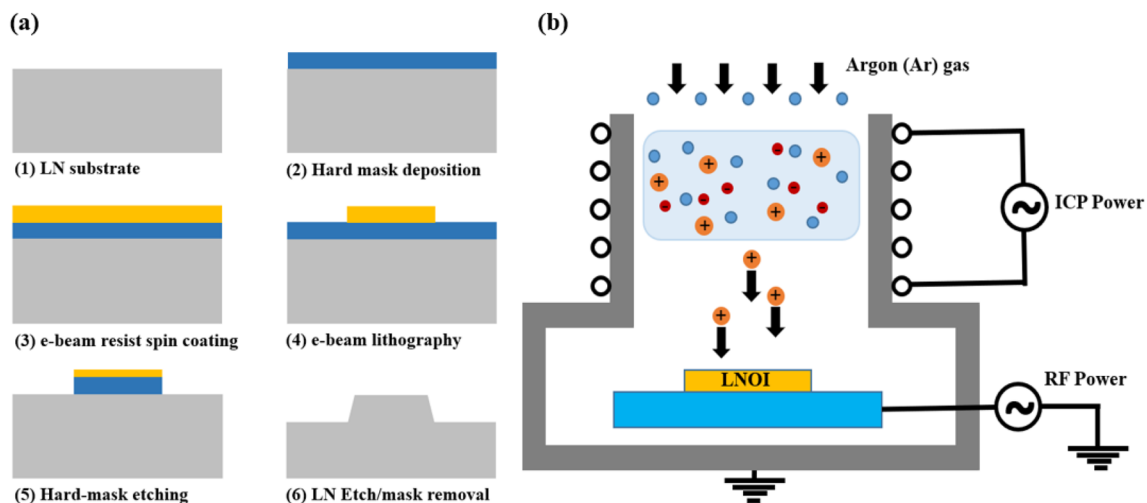


FIG. 1. Schematic representation of (a) fabrication process for patterning LN using the ICP RIE tool, resulting in non-vertical sidewalls on the etched structure, as shown in step (6). (b) Diagram of the ICP chamber used for LN etching. Argon gas is ionized under the influence of the ICP power. Ar⁺ ions are directed toward the LN substrate, controlled by the RF power.

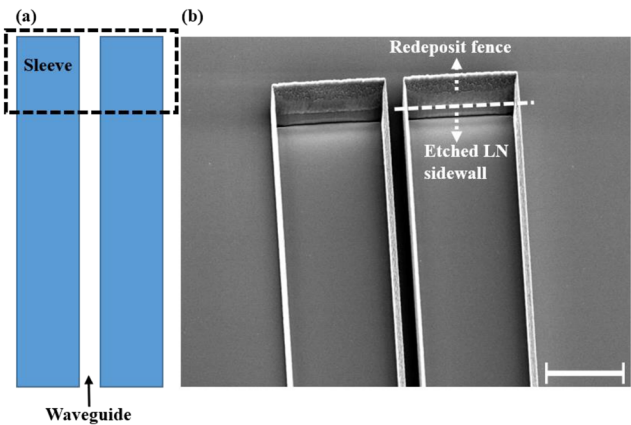


FIG. 2. (a) Schematic of the etched LN WG (shown by the bottom arrow) structure defined using sleeves (blue rectangular region) when employing positive-tone e-beam resist. (b) Top view SEM image of the region marked by a dashed rectangle in (a). A vertical fence of redeposit tops all the etched edges. The dashed white line on the right etched sleeve represents the boundary separating the etched LN from the vertical redeposit fence. The scale bar is 4 μm .

TABLE I. Etch chemistries used to define Cr and SiO₂ hard masks.

Parameter	Value	Value
Material	Cr	SiO ₂
RF power (W)	6	50
ICP power (W)	1200	600
Gas flow (SCCM)	Cl ₂ :42 He:50 O ₂ :8	C ₄ F ₈ :17
T (°C)	50	65
Pressure (mTorr)	12	1.2
Etch rate (nm/min)	55	130

The LN etch is performed with an ICP RIE tool (Oxford PlasmaLab 100). Before loading the LNOI samples, a series of CF₄/O₂ clean cycles are implemented to purge or burn out any SiO₂ residues or contaminants adhered to the chamber walls from previous etching runs. The samples are affixed to a blank silicon carrier wafer using a thermally conductive paste and transferred into the ICP chamber. A schematic of the ICP chamber is illustrated in Fig. 1(b). In this configuration, Ar gas is ionized as it passes through an inductively coupled coil under the influence of electromagnetic fields. The plasma density is controlled via the ICP power. The Ar⁺ ions are then directed toward the substrate, and their acceleration is regulated by the radio-frequency (RF) power (also called bias power). Based on our study and initial tests, we determined that the judicious selection of RF power dominantly influences the optimal etching performance. To this end, our first optimization is to systematically tune and optimize the RF power and discern the trend of the etched features. In this study, the values of ICP power (1500 W), pressure (5 mTorr), Ar gas flow-rate (20 SCCM), and chamber temperature (5 °C) were maintained at constant values. We note that due to the limited thermal conductivity of LN, the LNOI samples

are susceptible to damage if they are exposed to continuous, long-duration etch. To mitigate overheating of the LNOI samples, we break the total required etch into 1-min-long etch segments, each separated by 5-min-long cooldown periods. During the cooldown cycle, the samples are moved into the load lock (still maintained under vacuum), left in the load lock for 2 min, and then transferred back to the etch chamber for further etching. The total delay time between the previous and the next successive etch is about 5 min. This process is repeated until the desired etch depth is achieved.

After the LN etch, we remove the masking materials. For the organic resist mask, we use remover 1165; the Cr and the SiO₂ layers are stripped with chromium etchant and 6:1 buffered oxide (BOE) etchant, respectively, revealing the etched LN features and etch-material redeposit. An image taken with scanning electron microscopy (SEM) of the etched LN just after mask removal is shown in Fig. 2(b). Here, we see that all the etched edges are covered by vertical “fences” of redeposited material. Figure 2(a) sketches the waveguide and the sleeves used to define the waveguide due to the positive-tone resist.

We employ a wet cleaning process³² to remove the material redeposited on the LNOI during the ICP etching. The process is based on an RCA-1⁴⁵ solution (2:2:1 solution of NH₄OH, H₂O₂, and H₂O). The solution is constantly stirred during the cleaning using a magnetic stir bar. The samples are mounted in custom-designed boats that allow the sample to be oriented interchangeably at 0° and 90° relative to the stirring direction. The boat is immersed in the RCA-1 solution heated to 85 °C. The wet cleaning time duration in the RCA-1 solution is optimized to entirely remove the redeposition while avoiding damage to the LN WGs. The optimized duration is 15 min in both boat orientations, 0° and 90°. The total cleaning time is 30 min. Details of the wet-cleaning optimization are presented in Sec. III B.

To conduct a deeper and more precise analysis of the impact of the hard mask on WG properties, including etch depth and sidewall angles, we conducted a comprehensive analysis employing focused ion beam (FIB) milling in conjunction with scanning electron microscopy. We first deposit a metallic layer on the WG and then ion-mill the structure to create a 1 μm deep trench. This step enables detailed SEM imaging of the cross-section of the WG, unveiling both waveguide sidewall angle and etch depth.

III. RESULTS AND DISCUSSIONS

A. Influence of RF power

Our initial focus on optimizing the etching process involved systematically tuning the bias RF power of the ICP RIE process to etch LNOI samples. This parameter is pivotal in accelerating ions toward the substrate, thereby controlling the etch rate and etch depth. We vary RF power from 50 to 500 W while maintaining all other parameters at constant values. At an RF power of 50 (500) W, the etch rate for LN was 50 (107) nm/min. Examining the sidewall redeposition trends at these different RF power settings is crucial, as the redeposition pattern significantly influences the physical characteristics of the sidewall, as will be described in this section.

Figures 3(b)–3(e) illustrate the dependence of redeposit formation with varying RF power obtained in samples with a Cr

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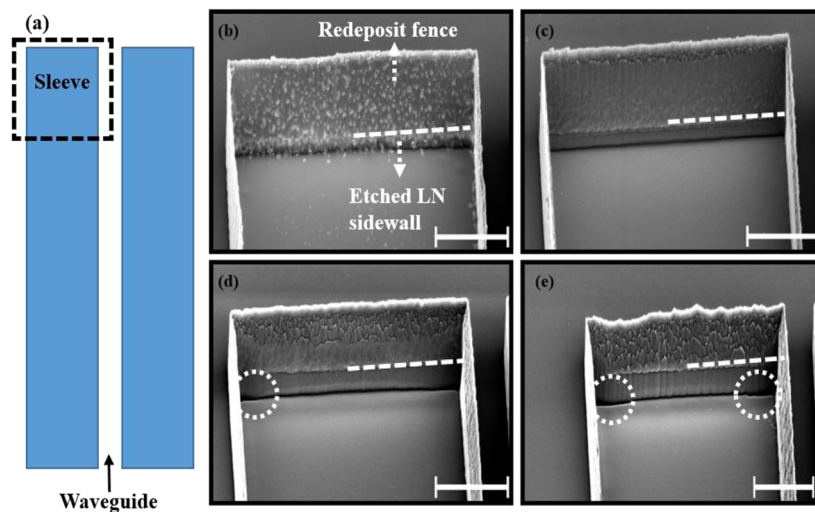


FIG. 3. (a) Schematic of the etched LN WG (shown by the bottom arrow) structure defined using sleeves (blue rectangular regions) employed for the positive-tone e-beam resist. The dashed box represents the top view of the region analyzed using SEM at RF powers of (b) 50 W, (c) 100 W, (d) 300 W, and (e) 500 W in samples using the Cr mask. The dashed white line represents the boundary separating the top surface of the etched LN film from the vertical fence of the redeposit. The dashed circles at the WG corners in (d) and (e) represent the damaged regions seen at high RF power. All scale bars are 1 μm .

mask. Figure 3(a) shows a sketch of the etched structure, and Figs. 3(b)–3(e) show the inner wall of the boundary between the unetched and etched regions. These figures show the redeposition along the short edge of the structure, but we find that the redeposition is formed consistently along all the edges, irrespective of the orientation. The redeposit formation phenomenon can be explained as follows: as the ions impinge upon the LNOI chip, they initiate the etching of the exposed LN. Concurrently, these ions also bombard the mask material, which at the beginning of the etch process comprises the resist. Therefore, the redeposit material consists of a blend of etched LN and resist. This redeposit begins to manifest as vertical fences on top of the sleeve edges, with a thickness of ≈ 40 nm. This phenomenon is also referred to as fencing.^{46–48} Notably, as depicted in Figs. 3(b)–3(e), the fencing formation persists regardless of the RF power setting. The morphology of the redeposition stands in contrast to trends in redeposit formation reported elsewhere.^{32,44} We believe this difference is attributed to the fact that the fabrication process might be intrinsic and unique to the ICP RIE etch tool. Another factor to consider is that the ICP RIE tool was not exclusively designated for LN etching. Instead, other materials are etched into this tool. As a result, etch residues may have remained on the chamber walls from etching different materials, despite thorough cleaning with CF_4/O_2 plasma after each ICP RIE etch. Nevertheless, we achieve a consistent replication of the etching results. We emphasize that the RF power needs to be optimized depending on the tool since the redeposit might be dependent on the LN etch tool, the tool usage history, and the materials that were etched on this tool.

In all of the SEM images depicted in Fig. 3, a free-standing, vertical fence of redeposit forms above the etched features. The dashed lines in Figs. 3(b)–3(e) mark the boundaries between the top surface of the etched LN film and the fence. The entire region below the

dashed line is the etched LN sidewall, while the region above the line is the vertical redeposit fence. It should be noted that this vertical fence seen in Figs. 2 and 3 should not be confused with the actual LN sidewall.

At a low RF power (≈ 50 W), the etched LN sidewall is entirely enveloped by the redeposit. This is illustrated in Fig. 3(b). As the RF power is incremented to 100 W, sections of the LN WG's sidewall become devoid of redeposit, as depicted in Fig. 3(c). A clear morphology distinction can be noted above and below the marked line at this RF setting, indicating that the redeposit is slowly removed from the LN sidewall. Further elevating the RF power to 300 W results in the complete elimination of redeposit from the LN sidewall, as seen in Fig. 3(d) (however, the redeposit fences above the WG remain in all the images). This behavior can be explained by considering the etch and redeposition rates, which depend on the impinging ion's energy.

As the ions bombard the LN, waste products are redeposited on the sidewalls, governed by the redeposition rate. If the etch rate can be made significantly higher than the redeposition rate by increasing the RF power, the sidewalls can be cleared of the redeposit during the etch step itself.⁴⁴ However, by optimizing the RF power, we can obtain redeposit-free sidewalls, but the free-standing vertical fence of redeposit atop the WG still remains. Therefore, a wet cleaning process is needed to remove this redeposit fence. In Sec. III B, we discuss the redeposit cleaning process.

Note that the LN etch rate increases with higher RF power, leading to deeper etch depths that can result in better optical-mode confinement. While higher RF power achieves redeposit-free sidewalls, the line-edge roughness of the free-standing vertical fence of redeposited material is transferred to the etched WG at RF powers beyond 300 W, which is seen as stripes along the WG's sidewall [Fig. 3(e)]. Therefore, it is critical to operate below this threshold.

At an RF power of 300 W, WG damage is observed, manifested in areas marked by dashed white circles in Figs. 3(d) and 3(e), where trenches/pits begin forming at the WG's edges. The damage becomes more pronounced at higher RF powers. For example, at an RF power of 500 W, the WG damage intensifies with the trenches widening. This study concludes that an optimum operating region for the RF power would be between 100 and 200 W (for our specific ICP RIE tool). This process underscored the importance of judiciously setting the RF power to achieve a delicate balance for controlling the etch depths, eliminating sidewall redeposition, and preventing damage to the WG.

B. Redeposit cleaning optimization

Redeposition removal through cleaning is a crucial step in obtaining low-optical-loss LN WGs. As mentioned previously, immediately after etching, the WGs are covered with significant amounts of redeposited material, as depicted in Fig. 4(a) (where no RF power optimization was done to remove the redeposit from the LN sidewalls). To remove the redeposit, it is necessary to develop a cleaning process. Our investigations show that an RCA-1a solution (NH_4OH , H_2O_2 , and H_2O in ratio 2:2:1) heated to 85°C was superior in removing the redeposit when compared to alternative chemical solutions such as Piranha (a mixture of H_2SO_4 and H_2O_2 in 3:1 ratio) or a related RCA-1b solution (NH_4OH , H_2O_2 , and H_2O in ratio 5:1:1). Here, “a” and “b” are used to identify the differences in the RCA-1 solution's composition. Furthermore, the methodology of the cleaning cycle itself must be optimized. This is because the free-standing fence of redeposited material produced during the etching process [shown in Fig. 4(a)] can collapse or fall over during cleaning, leaving debris across the WG region, as shown in Fig. 4(b).

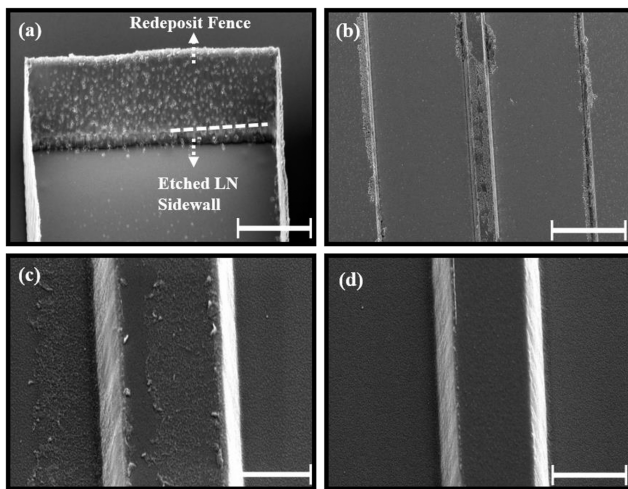


FIG. 4. SEM images of etched LN WG obtained (a) immediately after etching without any RF power optimization to remove sidewall redeposit, and (b) 10-min, (c) 15-min, and (d) 30-min total cleaning time. The dashed white line in (a) represents the boundary separating the top surface of the etched LN film from the vertical fence of the redeposit. Scale bars for (a) and (b) are $2\ \mu\text{m}$, and (c) and (d) are $400\ \text{nm}$.

The redeposit is removed upon longer cleaning times, as shown in Figs. 4(c) and 4(d). In our experiments, we had good results for redeposition removal by cleaning for at least 15 min in each 0° and 90° angle boat orientations. Whenever the orientation is changed, a fresh solution bath is prepared. Considering that the redeposit fences are formed along the shorter and longer edges of the rectangular sleeves, we find that changing the boat orientation helps to remove the redeposit. It is crucial to note that WGs free from redeposition are imperative to prevent light scattering during LN photonic device operation.³² While it may seem intuitive that extended cleaning durations would effectively remove redeposit, it is noteworthy that the process is not entirely straightforward and warrants discussion. Surprisingly, excessive cleaning can adversely affect the WGs in multiple aspects.

When we prolong the immersion of samples in the RCA-1 solution beyond the optimized 30-min (15 min each at 0° and 90° angles) cleaning time, we find that prolonged cleaning leads to chipping of the WG. This phenomenon is more pronounced along the sidewall edges [indicated by arrows in Fig. 5(a) with a magnified view of the damage in Fig. 5(b)]. The chipping is observed along the entire length of the WG. Likewise, instances are noted where more significant portions on the top of the WG are also peeling off, as illustrated in Fig. 5(c). This phenomenon adversely impacts the optical properties of the WG, leading to increased losses and the scattering of light out of the WG.

Another adverse consequence of prolonged cleaning is the change in the geometrical properties of the WG, including etch depth and sidewall angles, from the intended values. To illustrate this point, samples were intentionally left in the RCA-1a solution for 24 h. The cross-section of the sample cleaned for 24 h is depicted in Fig. 6(a), while that of the sample cleaned for 30 min is presented in Fig. 6(b). A marked contrast is apparent between the two. Figure 6(a) shows the WG cross-section has become asymmetric with unequal sidewall angles on either side. We believe that this asymmetry arises from the excessive cleaning that leads to differential etching of the

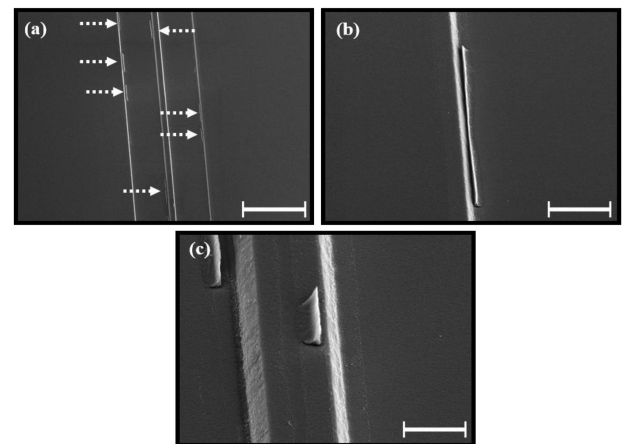


FIG. 5. Top view SEM images of etched LN WG after 50-min total cleaning time, which resulted in chipping at multiple points along the length of the WG indicated by white arrows in (a). Magnified view of (b) a chipped region and (c) peeling off at the top of the WG. Scale bars are (a) $5\ \mu\text{m}$, (b) $1\ \mu\text{m}$, and (c) $500\ \text{nm}$.

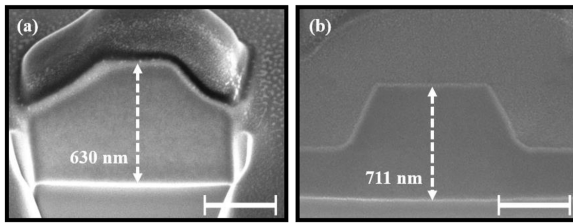


FIG. 6. Cross-sectional FIB milled/SEM images of etched LN WG obtained after (a) 24 h of cleaning (resulting in changes in WG geometry in both sidewall angles and total film thickness) and (b) 30 min of total cleaning time. The thick layers on top of the WGs are layers of 50 nm carbon and 200 nm platinum deposited to facilitate better contrast imaging after FIB milling. Scale bars are (a) 300 nm and (b) 500 nm.

various crystal surfaces of LN. Our observations also indicate that the sample subjected to excessive cleaning exhibited reduction in total LN film thickness (≈ 630 nm) compared to the as-obtained wafer from the vendor, which had an LN thickness of 700 ± 10 nm. In contrast, the sample cleaned for 30 min retained the same thickness and had equal sidewall angles of $\approx 70^\circ$, comparable to reports published elsewhere.¹⁶

Deviations in the WG geometry from the modeled values change the effective refractive index from the desired values. Also, uneven sidewall angles can induce asymmetry in the mode profile, resulting in scattering and coupling losses.⁴⁹ Such variations adversely impact the overall performance of the WG, affecting transmission efficiency and the capacity to confine and guide light effectively.

To achieve low-loss LNOI WGs, we must minimize sidewall roughness and eliminate sidewall redeposit that can induce scattering and loss.³² It is critically important to implement an effective cleaning protocol. To reduce dependence on the chemical cleaning process, one strategy is to address redeposition during the etching step, as suggested in Sec. III A. This simplifies the fabrication procedure and lowers the risk of damage during further handling of the WGs.

C. Soft mask vs hard mask

We initially utilized only the soft mask to transfer the EBL patterns from the resist to the LN layer. Using a soft mask streamlines the fabrication process economically and timewise, as numerous intermediate steps associated with a hard mask (such as hard mask deposition, its subsequent etching, and its later removal) can be avoided. However, a soft mask for pattern transfer to the LN layer proved itself to be non-ideal in our process. Figure 7(a) illustrates the SEM image of the LN etch profile obtained after the redeposition removal using the soft mask. The striated etch defects on the sidewalls of the WG are clearly seen in the image. These defects likely arise due to the degradation of the line-edge roughness of the soft mask during the ICP RIE etch. Such defects lead to higher optical losses and are highly undesirable. Regardless of how the RF power was tuned, these striations persisted and proved challenging to avoid by only varying the RF power. After discovering the soft mask produced striated WGs, we switched to a Cr hard mask. We etched

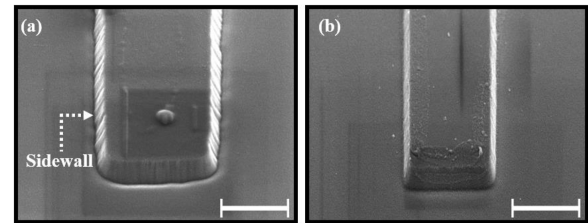


FIG. 7. Top view of SEM images of etched LN WG obtained using (a) a soft mask resulting in stripes or striations along the sidewall and (b) a Cr hard mask. The particles on the WG are the remnants of the redeposit. Dark rectangular boxes are artifacts from the SEM imaging. Scale bars are (a) 500 nm and (b) 700 nm.

a WG test structure of similar thickness, maintaining consistent etching parameters as used with the soft mask.

The sidewall profile exhibits a marked improvement using the Cr hard mask, as illustrated in Fig. 7(b). This underlines the critical importance of judiciously selecting a mask material. In our approach, we chose to use a hard mask over a soft (organic e-beam resist) mask. We note that our intention is not to assert that a soft mask cannot be employed for efficient etch transfer of the pattern, and in fact, previous reports suggest their usage.³² However, it should be emphasized that the redeposition formation observed in our process differs significantly from what has been reported in existing literature, making it impractical to use a soft mask if low optical losses in WG are desired. In Sec. III D, we compare two different hard-mask materials: Cr and SiO₂.

D. Hard masks comparison

Section III C establishes that a hard mask produces a better etch than a soft mask. This section now compares the LN etch profiles obtained using metallic (Cr) and dielectric (SiO₂) hard masks with identical LN etch processing parameters: RF power (150 W), ICP power (1500 W), pressure (5 mTorr), gas flow (20 SCCM), and temperature (5 °C). Figure 8 presents the SEM images of the etched profiles achieved with Cr [Fig. 8(a)] and SiO₂ [Fig. 8(b)] hard masks. We observe a difference in etch quality between the tested hard masks.

In particular, we notice that the etched features using the Cr mask exhibit granular features along the sidewall compared to the

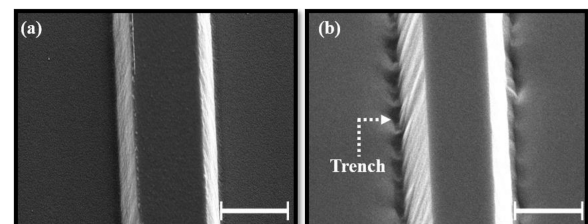


FIG. 8. SEM images of the top angled view of etched LN WG obtained using (a) a Cr hard mask and (b) a SiO₂ hard mask. Trenches are seen at the bottom of the WG, along its sidewall edges, as indicated by the arrow on (b). Scale bars are (a) 500 nm and (b) 700 nm.

striated sidewalls obtained using the SiO₂ mask. The granular features on etched LN WG can be attributed to the chromium hard mask's granular structure and related line-edge roughness. Cr is commonly used in its polycrystalline form, characterized by crystalline grains. This grain size depends on the deposition process parameters, including the deposition rate and chamber pressure. In our process, a 50 nm thick layer of Cr is deposited using an e-beam evaporator at a 0.3 Å/s deposition rate and 5×10^{-6} Torr chamber pressure. The related Cr grain size is ≈ 55 nm, as characterized by an SEM measurement. These grain-size-related defects are then transferred to the underlying layers during subsequent etching. In contrast, SiO₂ is often found in an amorphous or glassy form, lacking a well-defined grain size, resulting in smoother etched sidewall profiles. However, our etch results showed striations on the sidewalls using the SiO₂ hard mask, resulting in more significant sidewall roughness compared to using the Cr hard mask.

Another noticeable feature in the image where the SiO₂ hard mask is used is “trenching” along the base of the sidewalls [Fig. 8(b)]. “Trenching” refers to the irregular pit patterns observed along the length of the sidewalls. We speculate this phenomenon unfolds during the LN etching step while using SiO₂ as the hard mask. During the etch, the positive Ar ions impinge in a direction normal to the sample surface, inducing the etch. Given the insulating properties of both SiO₂ and LN, charges can become trapped and build-up, resulting in local electric fields that can deflect a portion of the impinging ions, thus leading to uneven bombardment. This leads to trenching observed at the edges of the WG. Surface charging leading to various damages, such as pitting and trenching in etched WGs using SiO₂ masks, have also been reported elsewhere.^{50–52} In this sample, we used a 10 nm thick Cr layer deposited over SiO₂ to enhance charge dissipation and reduce trenching, but this Cr layer did not prevent trenching. The trenching may contribute to striated features, causing a rough sidewall when using the SiO₂ mask. We note that the samples utilizing the Cr hard mask [Fig. 8(a)] do not display any trenching.

Further precise analyses of the impact of the hard mask material on the WG properties, including etch depths and sidewall angles, are revealed using FIB milling and are illustrated in Fig. 9. Aside from the observed trenches highlighted in dotted circles in Fig. 9(b), both etched WGs using Cr and SiO₂ masks exhibit comparable sidewall angles and etch depths. If a SiO₂ hard mask needs to be used, another

potential remedy to resolve the trenching issue would be to operate using modified etch parameters that can minimize their random deflection and reduce the occurrence of trenching.⁵¹ Having studied and compared the Cr and SiO₂ hard masks, we decided to use the Cr hard mask to fabricate WGs for optical loss characterization.

E. Optical characterization

We investigated the optical performance of the etched LNOI WGs. We used the process detailed above to realize rib LNOI WGs using a Cr mask. We optimized ICP parameters and redeposition cleaning time with the parameters given in Table II. The dimensions of the WGs are a top width (w) of 600 nm, a sidewall angle (θ) of 70°, and an etch depth (h_{rib}) of 350 nm over an LN slab of height (h_{slab}) of 350 nm, which is sketched in Fig. 10(a). These dimensions are chosen so that the WGs are in single mode at a wavelength of 1550 nm, which we confirmed using a numerical mode solver. The optical loss characterization is performed on eight identical, 4.5 mm long, straight LN WGs that we patterned and etched on the same LNOI chip using a Cr hard mask.

Before launching light into the WGs, we cladded the chip consisting of the eight WGs with a 2 μm thick SiO₂ layer using PECVD. We then mechanically polished the end facets of this chip using lapping films. Thus, the eight WGs were polished together, resulting

TABLE II. Optimized LNOI WGs fabrication parameters for optical loss characterization.

Parameter	Value
Mask material/thickness (nm)	Cr/50
RF power (W)	150
ICP power (W)	1500
Gas flow (SCCM)	20
T (°C)	5
Pressure (mTorr)	5
Total redeposition cleaning time (min)	30

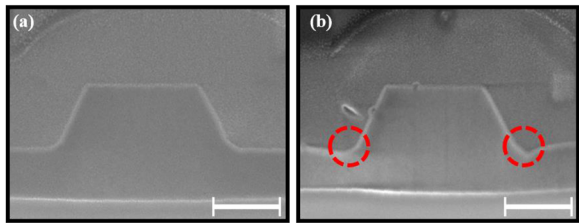


FIG. 9. Cross-sectional FIB milled/SEM images of etched LN WG obtained using (a) Cr and (b) SiO₂ hard masks. Red circles represent trenching issues using a SiO₂ mask. The two minor dot-like defects in (b) are the artifacts due to FIB milling. The scale bars for (a) and (b) are 400 nm.

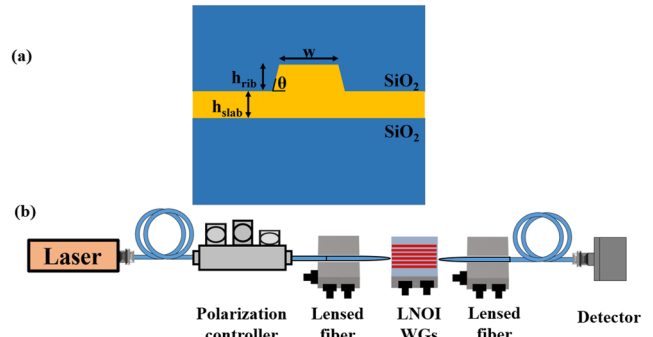


FIG. 10. (a) Schematic cross-section of x-cut LNOI WG with dimension: top width (w) is 600 nm, sidewall angle (θ) is 70°, rib height (h_{rib}) is 350 nm, and slab height (h_{slab}) is 350 nm, which is extracted from FIB/SEM measurements. (b) Schematic of the experimental set-up for measuring optical loss at 1550 nm laser input in eight LNOI WGs of length 4.5 mm. The lensed fibers (LNOI chip) are placed on an xyz (xy) stage.

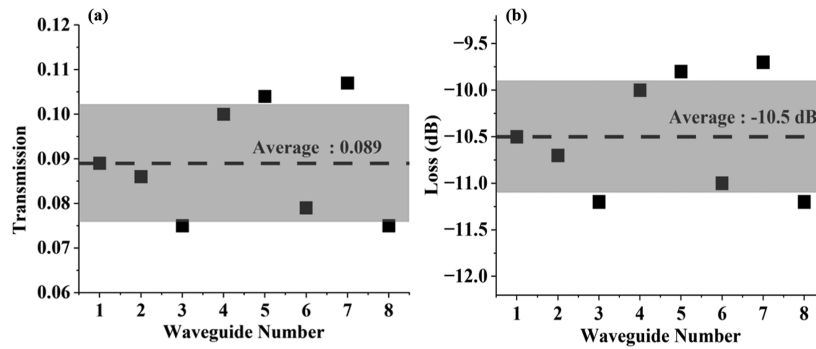


FIG. 11. Results of the optical loss measurements performed on eight identical LNOI WGs of length 4.5 mm on a single chip. The shaded region represents the standard deviation of the measured values. The dashed line represents experimentally obtained average values of (a) 0.089 transmission (output power/input power) and (b) -10.5 dB total loss (due to coupling and propagation losses) across the eight WGs.

in identical lengths. Light from a fiber-coupled 1550 nm wavelength laser is coupled into and out of the polished facets of the WG using lensed optical fibers with a mode field diameter of 2.5 μm , as illustrated in Fig. 10(b). The test chip's facets did not have anti-reflection coatings. The sample is translated using a mechanical stage, and the output power collected into the lensed fiber at each of the eight WGs is recorded. Figure 11(a) plots the transmission variation, while Fig. 11(b) shows the equivalent loss values across the eight WGs. The transmission (loss) ranged from 0.075 (-11.2 dB) to 0.107 (-9.7 dB). The average transmission is $(8.9 \pm 1.3)\%$, corresponding to (-10.5 ± 0.6) dB insertion loss, including propagation and coupling losses.

Our total loss measurement values are comparable with the values found in the literature, as tabulated in Table III.^{53–55} As an approximation, we provide an upper-bound estimation of waveguide propagation losses. For LNOI butt couplers without tapers (the case relevant to this work), a typical insertion loss of light from a lensed fiber ranges from 4.8 to 10 dB/facet.^{25,27,53,56} To get an upper-bound estimate of the propagation loss, we approximate the coupling losses per facet to be equal to their low-bound value (4.8 dB). This gives us ~ 2 dB/cm propagation loss. We would like to comment that this is a very rough estimate of the propagation losses due to our inability to separate propagation and coupling losses. Notwithstanding, the total losses we measured are consistent with the literature, which provides evidence that our fabrication recipe results in WGs that are useful for optical experiments. However, more accurate values of propagation losses could be

further characterized by employing cut-back method or by Q-factor measurements.^{55,57}

IV. SUMMARY AND CONCLUSIONS

In summary, we have presented LNOI etching protocols developed using the NIST NanoFab shared clean room facility. We have investigated the influence of mask material, RF power etch parameter, and the redeposition cleaning process on the quality of etched WGs. Our results for etching LNOI show a distinctive redeposition morphology consisting of a free-standing vertical fence that differs from prior reports. We present a detailed comparison of etched WGs employing various mask materials. Our proposed fabrication process's primary objective is to mitigate the impact of the redeposit and minimize edge roughness on the LN WGs and, with that, achieve small propagation losses. Emphasizing the critical role of an appropriate cleaning protocol in removing the redeposit, we demonstrate how deviations can adversely affect the WG quality. Using the established recipe, we demonstrated low-loss optical WGs with total (combined propagation and coupling) loss values consistent with the literature. We believe the fabrication processes and lessons we have learned will clarify the path to realizing low-loss LNOI WGs. Our work will serve as a guide that advances the field of LNOI research and makes the fabrication of LNOI WGs more accessible, particularly to those using shared clean-room facilities.

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TABLE III. Comparison of total loss (due to coupling and propagation losses) values reported in the literature at 1550 nm input for etched LNOI WGs of similar length.

References	Sample length (mm)	Loss (dB)
53	3	-10.2
This work	4.5	-10.5
54	5	-15
55	6	-15

Certain commercial equipment, materials, or computational software are identified in this paper to specify device fabrication, the experimental procedure, and data analysis adequately. Such identification is not intended to imply endorsement by the National Institute of Standards and Technology, nor is it intended to imply that the equipment, material, or software identified are necessarily the best available.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

CH. S. S. Pavan Kumar: Investigation (lead); Writing – original draft (lead); Writing – review & editing (equal). **Nikolai N. Klimov:** Methodology (lead); Supervision (equal); Writing – review & editing (equal). **Paulina S. Kuo:** Conceptualization (lead); Methodology (supporting); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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